



High-strength ultrafast laser micro-welding of sapphire and silicon



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Abstract

A direct joining of sapphire and silicon with significantly different coefficients of thermal expansion (CTEs) was successfully realized via ultrafast laser welding. The microstructure, mechanical properties, and fracture modes of sapphire/Si joints were systematically analyzed. The typical microstructure of the joint consists of three regions, including an area of nanoscale α -Al₂O₃ particles embedded in the amorphous Al-O-Si phases, a transition region with locally recrystallized Si, and a region of amorphous Si adjacent to the Si substrate. By optimizing the laser parameters (5.74W laser power, 50mm/s scanning speed, 800 fs pulse width), defect-free joints were obtained. The optimized joints exhibited a peak shear strength of 38.4MPa, which is the highest value compared to state-of-the-art Al₂O₃/Si joints. Such a high strength value is attributed to the formation of nanoscale α -Al₂O₃ particles embedded in the amorphous phases, which increases the joint resilience. We hypothesize that our prepared

high-quality sapphire/Si joint and the ultrafast laser welding method have great potential applications in high-performance MEMS systems or microfluidic devices.



Keywords

Ultrafast laser; Sapphire; Monocrystalline silicon; Microstructure; Shear strength

1. Introduction

With the development of miniaturization and multifunctionality, the service conditions of micro/nano devices have become increasingly demanding, such as high requirements of precision and stability for the micro-chips, especially for the highly integrated heterogeneous material packaging and joining for the chips. Si and transparent materials like glasses are mainly used as substrates for semiconductor devices and integrated circuits [1,2]. For example, in Micro-Electro-Mechanical Systems (MEMS), optical window materials are commonly used to encapsulate the functional units integrated on silicon substrates to maintain optical accessibility [3]. Nowadays, glass has excellent stability and is widely used as a packaging material for MEMS devices. However, it still presents limitations in terms of light transmittance in specific optical bands and heat-dissipation capability [4]. As a new generation packaging material, sapphire combines optical transparency with exceptional properties such as corrosion resistance, high-temperature stability, extreme hardness, and superior thermal conductivity. Compared to glass, sapphire offers more stable optical transmittance, hardness, and chemical resistance. Based on the above properties, achieving high-quality and high-precision wafer-level bonding between sapphire and Si has shown significant potential in advancing MEMS devices, particularly in microfluidic chips [5].

Due to the large physical property gap between sapphire and Si, it is difficult to form an effective joint between sapphire and Si [6]. Diffusion bonding [7,8], brazing [9,10], and adhesive bonding [11] with intermediate layers are typically used for joining sapphire or Si. These methods typically require the introduction of metallic interlayers, followed by atmosphere-controlled

furnace heating/cooling to promote interfacial diffusion and chemical reactions. However, the introduction of interlayers may complicate the precision control of the whole joining process, which is critical for MEMS devices. In addition, diffusion bonding and brazing involve heating and holding processes, which not only complicate the process but also result in low joining efficiency and large residual stress at the interface of the joint. Therefore, there is a demand to develop a direct joining method for sapphire and Si. It has been reported that plasma activation technology could realize the direct bonding of sapphire/Si and Si/Si [[12], [13], [14]]. However, plasma activation technology requires the surface to be activated with argon (Ar) gas before joining, and then the whole assembly needs to be post-treated at a high temperature after joining, rendering the process complex and expensive. At the same time, MEMS are highly temperature-sensitive, and the high-temperature environments may lead to performance degradation or failure. Therefore, there is a demand to develop a direct bonding method for sapphire/Si with better efficiency, high quality, and precision.

Over the past two decades, ultrafast lasers have developed rapidly, offering high peak power, ultra-short pulse width (10^{-12} s to 10^{-15} s), and micrometer-scale processing precision. Under the irradiation of ultrafast lasers, transparent materials undergo nonlinear photon absorption processes, which may include multiphoton ionization, tunneling ionization, and avalanche ionization, establishing the possibility of ultrafast laser welding with a minimal heat-affected zone (HAZ) [15,16]. Ultrafast laser welding of glasses and Si was first reported in 2006; Takayuki et al. [17] achieved a glass/Si joint with a strength of 3.74MPa by ultrafast laser welding. Subsequent studies have further demonstrated the feasibility of ultrafast laser bonding for homogeneous and heterogeneous material combinations, such as glass/metal [18,19], ceramic/metal [[20], [21], [22]], and ceramic/ceramic [23,24]. But only a few reports are for semiconductor bonding. Chambonneau et al. [25] achieved Si/Si bonding by combining the resulting exalted absorption with filament relocation during ultrafast laser irradiation, and the joint shear strengths of ~4MPa were obtained with 21 nm thick gold nanolayers. Miyamoto et al. [26] utilized high repetition rate picosecond (ps) lasers to suppress the violent evaporation with the aid of diffusion and convective flow between Si and glass, realizing a high joining strength of 45MPa. These studies demonstrate that ultrafast lasers can weld dissimilar materials with silicon. However, few research studies have systematically investigated sapphire/Si joining. Hu et al. [6] developed a metallic nanoparticles(NPs) thin film assisted plasmonic micro-welding technology, where an Ag NPs layer between sapphire and Si facilitated plasmon-enhanced energy coupling under laser irradiation. The joint had a shear strength of 5.4MPa and excellent impermeability even after thermal shocking (-40°C – 120°C). Hu et al. [27] similarly achieved joining of sapphire and Si by an intermediate oxide layer (SiO_2), attaining a joint strength of 10.7MPa while demonstrating high resistance to both acids and alkalis (pH 2 to pH 12). These studies demonstrate the possibility of utilizing ultrafast lasers for the joining between sapphire and Si using interlayers,

combining precision and minimal HAZs. However, interlayer-dependent methods increase process complexity and manufacturing costs. Therefore, the direct joining of sapphire and Si via ultrafast laser needs to be further investigated. In addition, the effect of processing parameters on the joint microstructure and strength remains unclear, and the strength of the joint needs to be further increased.

In this paper, direct joining of sapphire and Si via ultrafast laser was realized without introducing any interlayer. The process yielded the distinct morphology of a smooth joint. Furthermore, the influence of laser process parameters on the microstructures and shear strength of the sapphire/Si joint was analyzed. Through laser parameter optimization, a high shear strength of 38.7MPa was successfully achieved. This study aims to propose a potential strategy for MEMS device packaging by comprehensively studying the ultrafast laser bonding mechanisms between sapphire and Si.

2. Materials and methods

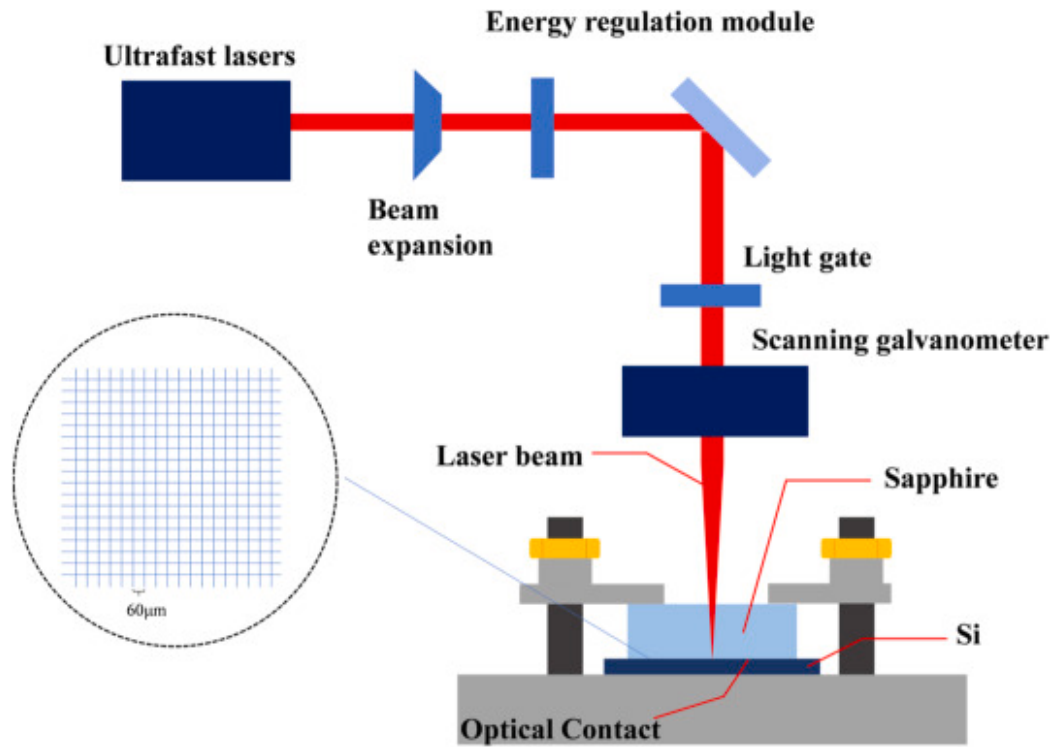
2.1. Materials

The sapphire used in the present study was provided by Hefei Yuanjing Crystal Co. Ltd. It was cut into 7mm×7mm×3mm, and the upper and lower surfaces were polished to transparency before the experiment. Another base material, monocrystalline silicon (Si), was provided by Kaihua Jingxin Electronics Co. Ltd. The Si substrate was cut into 10mm×10mm×0.625mm, and the faying surface was polished with diamond spray. The sapphire and Si were ultrasonically cleaned in acetone for 15min to achieve surface cleanliness meeting optical contact standards in joining.

2.2. Ultrafast laser welding process

[Fig. 1](#) shows the schematic diagram of the equipment setup for the ultrafast laser joining and the fixture to fix the sample. The ultrafast laser originates from Light Conversion, emitting 280 fs - 2 ps pulses with 1030nm wavelength, 1 kHz - 1 MHz pulse repetition rate, and 40W maximum average power. Correspondingly, the fixture can adjust the cover plate to compress the sample to achieve optical contact [\[28,29\]](#). During joining, the focused laser beam scans at the interface of sapphire/Fe-36Ni in an area of 4mm×4mm, following a pattern of cross lines, as shown in [Fig. 1](#), with a scanning interval of 60μm by an x-y galvo and output powers ranging from 3.11W to 13.74W, pulse width ranging from 268 fs to 1600 fs and scanning speeds ranging from 25mm/s to 150mm/s. Based on the laser spot size of 45μm, 60μm was selected as the baseline to ensure sufficient overlap for a

continuous texture. Under these conditions, the distinction between the unprocessed and processed regions becomes clearly observable while the shear strength is maintained.



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Fig. 1. Schematic of the equipment for ultrafast laser welding of sapphire/Si.

The laser energy corresponding to the different processing parameters in the present study is standardized by peak power density per unit area based on Eq. (1).

$$E_p = \frac{Df}{v} \times \frac{4E}{\pi D^2} \quad (1)$$

Where E_p is the peak power density, E is the single pulse peak power and D is the focused laser spot diameter, v is the laser

scanning speed, and f is the laser repetition frequency. The single pulse peak power E is calculated according to Eq. (2).

$$E = \frac{P}{f \cdot t_p} \quad (2)$$

Where P is the laser power, f is the laser repetition frequency, t_p is the laser pulse width. It can be known from Eqs. (1), (2) that the peak power density per unit area increases with the increase of laser output power, the decrease of laser scanning speed, and the increase of laser pulse width, respectively.

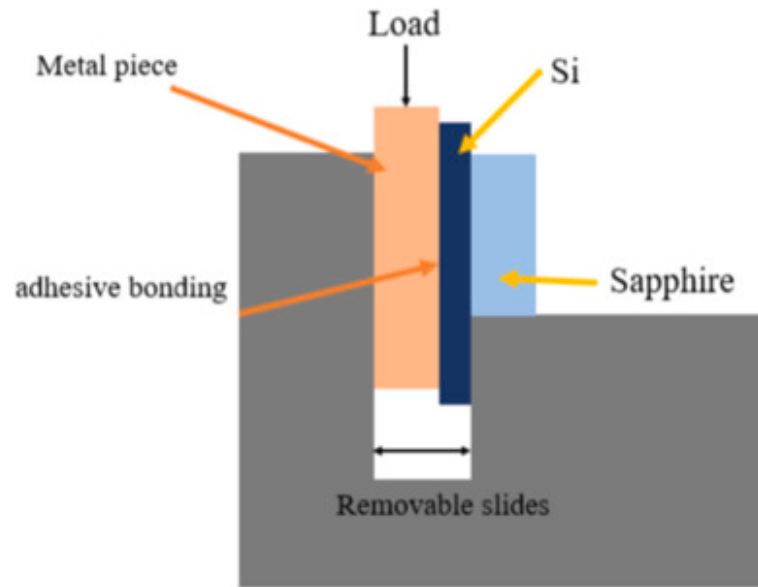
2.3. Analysis of microstructure and mechanical properties of the joint

A confocal laser scanning microscope (Olympus OLS4100) was used to observe the macroscopic morphology of the welding joints. The microstructure of the joint was analyzed by a scanning electron microscope (SEM, Quanta 650, FEI) with energy dispersive spectroscopy (EDS, AMETEK). Then the sample was prepared by focused ion beam (FIB) and the phase composition was determined by transmission electron microscopy (TEM).

The shear test was carried out at room temperature using the electronic universal testing machine (QBD-100 100kN) with a loading velocity of 0.5mm/min. The shear strength was calculated by dividing the maximum force during the shear test by the area of the scanning region, and subsequently calculating an average value based on at least three samples tested for each experimental parameter. As shown in Fig. 2, due to the fragility of Si, it was adhesive on the metal piece. The formula for calculating shear strength of the joint is shown in Eq. (3):

$$\tau = \frac{F_{\max}}{S} \quad (3)$$

where τ is the shear strength of the joint (MPa); $F_{\max}$ is the maximum fracture load in the shear test (N); S is the preset laser scanning area (mm²). After the shear test, SEM is used to analyze the fracture morphologies of sapphire and Si.



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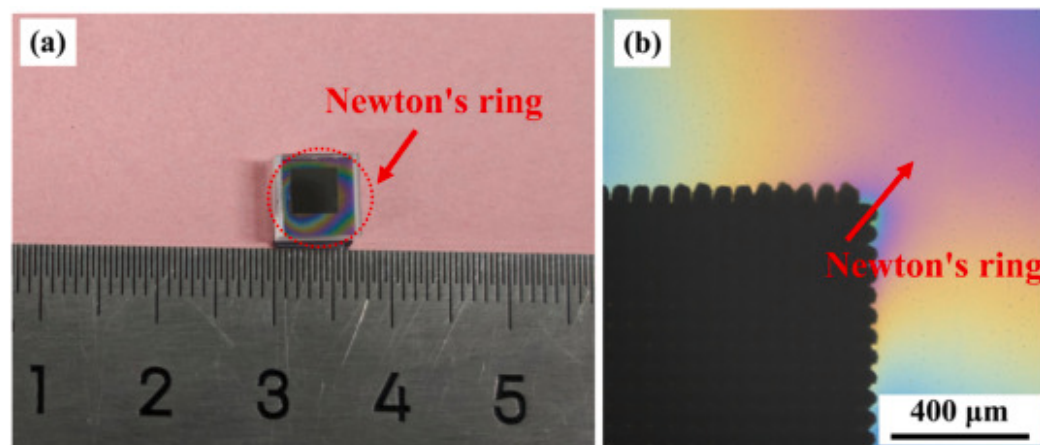
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Fig. 2. The schematic of the shear strength test.

3. Results and discussion

3.1. Typical microstructure of sapphire/Si joint

The macrostructure of the sapphire/Si joint is illustrated in Fig. 3. Color streaks from the formed Newton's ring can be observed in Fig. 3(a), indicating that the sapphire/Si joint interface is in optical contact, thereby confirming a successful joining between sapphire and Si [30]. In addition, as shown in Fig. 3(b), uniform and dense dark periodic welds without macrocracks, voids, or other defects in the edges can be observed under the irradiation of laser pulses, indicating the preliminary feasibility of sapphire/Si joining.

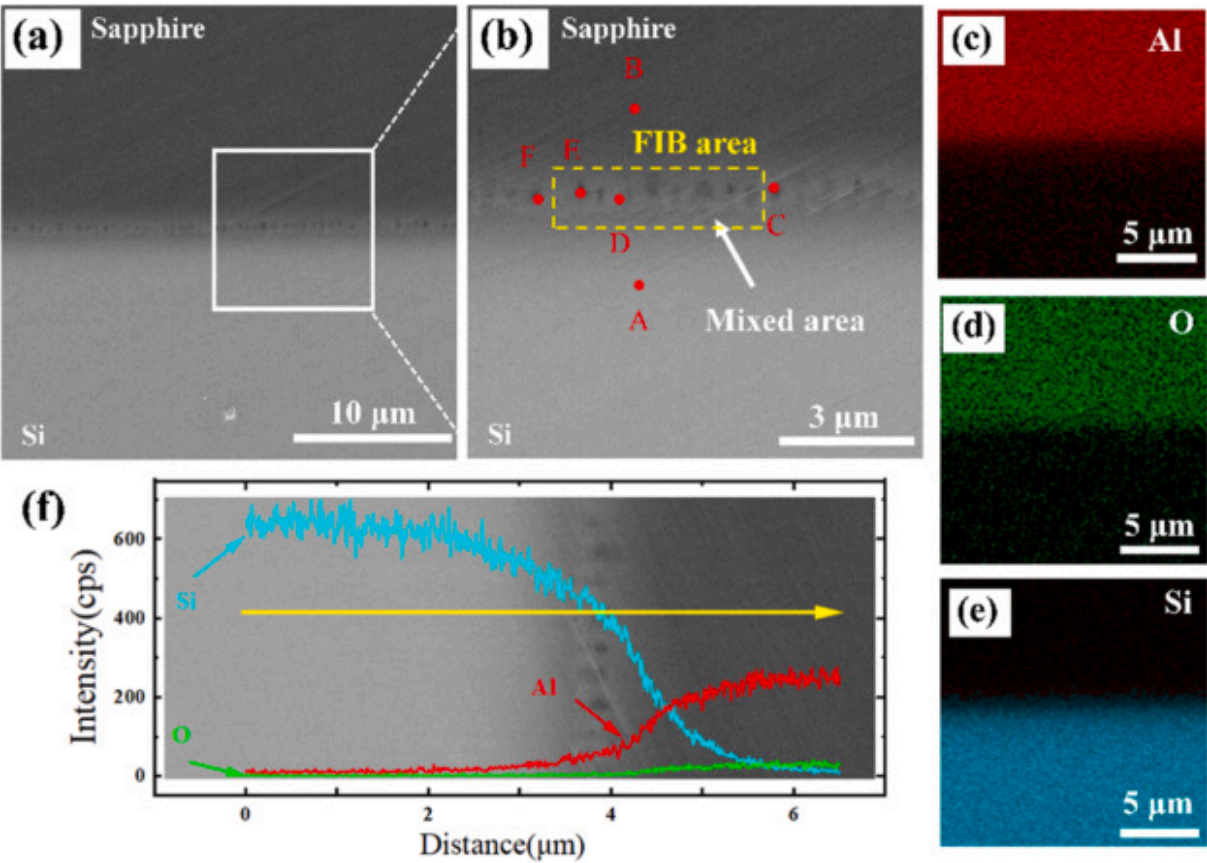


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Fig. 3. (a) The typical macroscopic morphology of sapphire/Si joint; (b) The edge morphology of sapphire/Si joint.

The typical microstructure of the obtained sapphire/Si joint welded by fs lasers with 800 fs laser width is shown in Fig. 4. As shown in Fig. 4(a) and (b), it can be seen that sapphire and Si are well joined by ultrafast lasers without any voids or microcracks at the interface. Specifically, a continuous and smooth joint with a thickness of $2.2\mu\text{m}$ is formed, and dark spot-like phases with nanoscale sizes are also formed in the joint domain. The surface element distribution results in Fig. 4(c–e) show that the three elements that Al, O, and Si, diffuse mutually at the interface. The line scan result shown in Fig. 4(f) indicates a continuous gradient transition of the elements at the interface, with a homogenized mixture of Al, O, and Si atoms. In addition, Table 1 shows elemental composition at different points in Fig. 4(b), and the results do not follow a clear pattern. This interface may form the amorphous regions during the recrystallization process of sapphire and Si, leading to an uneven distribution of elements. In order to further verify the above speculation and determine the phase composition, the FIB area in the marked zone of Fig. 4(b) was selected, and TEM characterization was conducted.



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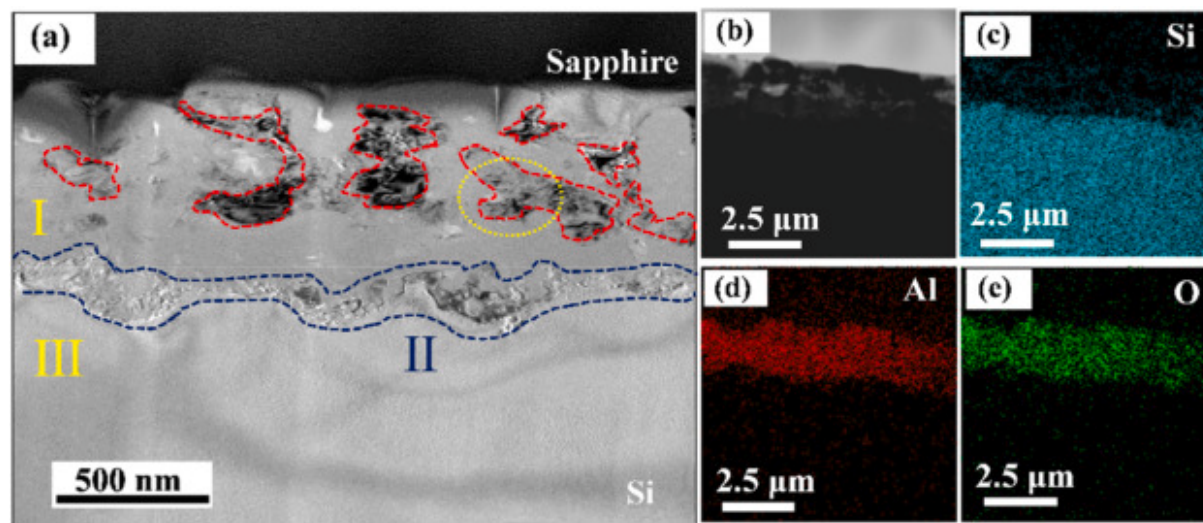
Fig. 4. Typical microstructure of smooth sapphire/Si joint joining by ultrafast laser with 5.74W, 800 fs,50mm/s: (a, b) Typical microstructure of the joint; (c–f) the EDS elements distribution.

Table 1. The EDS results of the points marked in the joint shown in Fig. 4 (at. %).

Point	O	Al	Si	Possible Phases
A	2.94	0.90	96.16	Si substrate

Point	O	Al	Si	Possible Phases
B	49.28	47.67	3.05	Al_2O_3 substrate
C	34.77	28.02	37.22	$\text{Al}_2\text{O}_3(\text{Si})$
D	30.46	23.64	45.90	$\text{Si}(\text{Al}_2\text{O}_3)$
E	39.71	35.01	25.27	$\text{Al}_2\text{O}_3(\text{Si})$
F	13.92	11.91	74.17	$\text{Si}(\text{Al}_2\text{O}_3)$

Fig. 5 (a) shows the bright field image of the selected FIB area of sapphire/Si joint, and the entire area can be divided into three regions, which include region I corresponding to the mixed area of the joint, containing the continuous dark matters; region II representing the transition region between the joint and the Si substrate; and region III corresponding to the Si substrate. To further analyze the structure of the dark spot-like phase and the mixed area, the area marked in a circle within region I was selected for High-Resolution Transmission Electron Microscopy (HRTEM) and selected-area electron diffraction (SAED) analysis. Fig. 5(b–e) show the surface element distribution results of the joint.

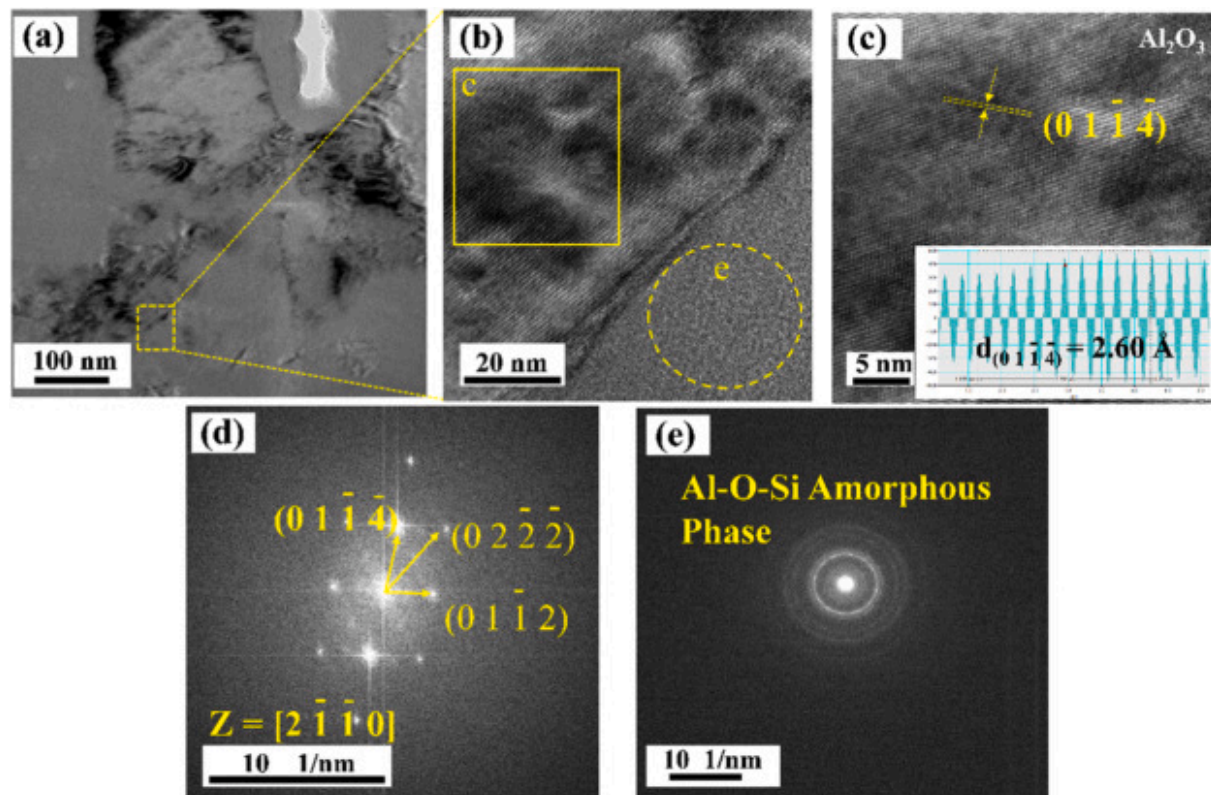


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Fig. 5. TEM analysis at the interface of sapphire/Si joint: (a) Bright field image of sapphire/Si joint domain; (b–e) EDS element distribution.

Fig. 6(a) shows the dark matter of the mixed area in Fig. 5(a). According to the HRTEM image in region c (Fig. 6(b)), it can be seen that the interplanar spacing of $2.60\text{\AA}$ corresponds to the (2 1 -1 4) plane (PDF#04-004-5291) of Al_2O_3 in Fig. 6(c). Meanwhile, the SAED pattern result in Fig. 6(d) by FFT also confirms the hexagonal Al_2O_3 , with crystal orientations identified as (0 1 -1 -4), (0 2 -2 -2), and (0 1 -1 2). The crystal structure of the dark spot-like phase remains consistent with that of the sapphire substrate. In contrast, region e (Fig. 6(e)) exhibits structural disorder characteristic of the distinct amorphous phase, and the EDS results in Fig. 5(c–e) confirm that this region represents an Al-O-Si amorphous phase. The formation of this amorphous phase is attributable to the extreme cooling rates and rapid crystallization under the influence of the ultrafast laser. From the above analysis, it can be indicated that the dark spot-like phase is an Al_2O_3 particle, and it is surrounded by the Al-O-Si amorphous phase in the joint due to the treatment of the ultrafast laser.



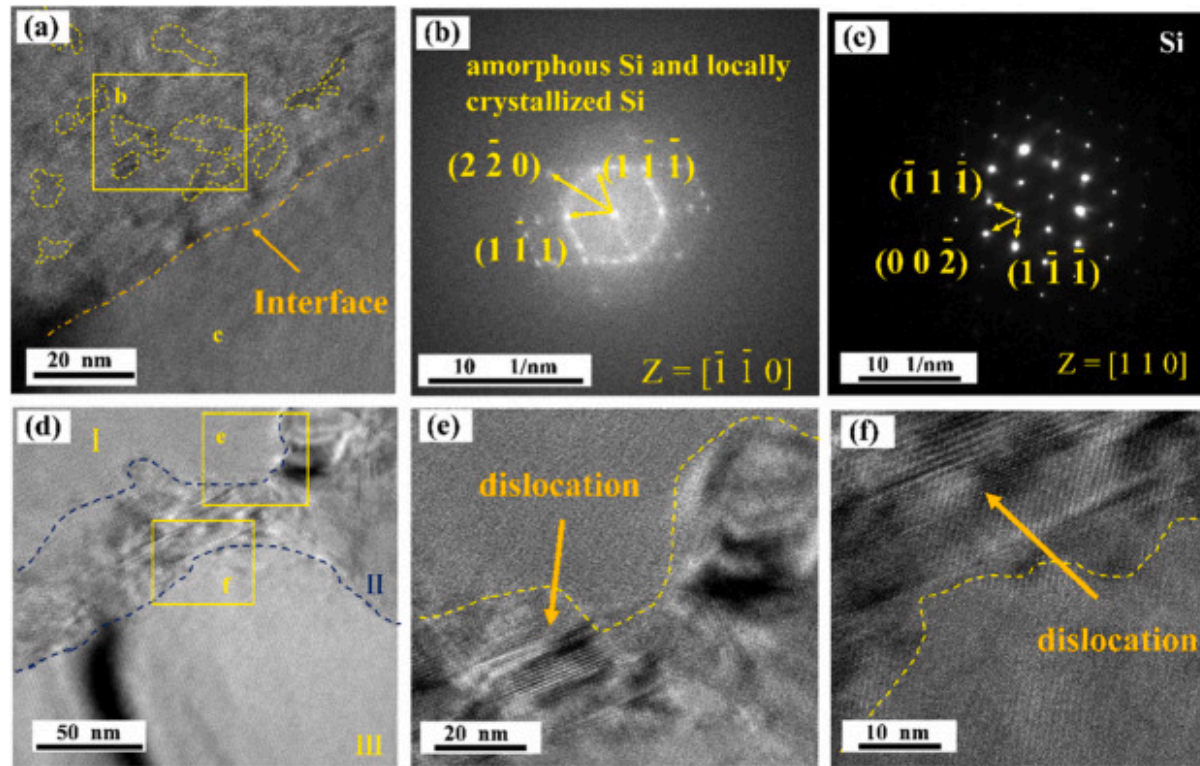
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Fig. 6. TEM analysis at the interface of sapphire/Si joint: (a) magnified image of the bright field image in Fig. 5(a); (b) the magnified bright field image of the marked zone in Fig. 6(a); (c) HRTEM image of the region c marked in Fig. 6(b); (d) SAED pattern results by FFT of Fig. 6(c); (e) SAED pattern of the region e in Fig. 6(b).

Fig. 7(a) shows the magnified view of the transition region II marked in Fig. 5(a). Fig. 7(b) reveals a mixed structure comprising both crystalline and amorphous phases. The SAED pattern in Fig. 7(b) confirms the presence of tetragonal Si (PDF#04-006-2591) in this region, with crystal orientations identified as (0-2 -0), (1 -1 1), and (1 -1 -1). As shown in Fig. 7(c), the SEAD pattern corresponds to the (0 0-2), (1 -1 -1), and (-1 1 -1) crystal planes of tetragonal Si (PDF#04-006-2591), indicating that region III is the Si substrate. Additionally, Fig. 7(d) shows another part within the transition region II from Fig. 5(a). From Fig. 7(e) and (f), it can be seen that some dislocations are observed in the mixed crystalline-amorphous Si regions. These dislocations may cause the

concentration of stress in the transition region II, possibly leading to fracture initiation at the Si substrate. This fracture behavior will be further discussed in the subsequent fracture surface analysis.



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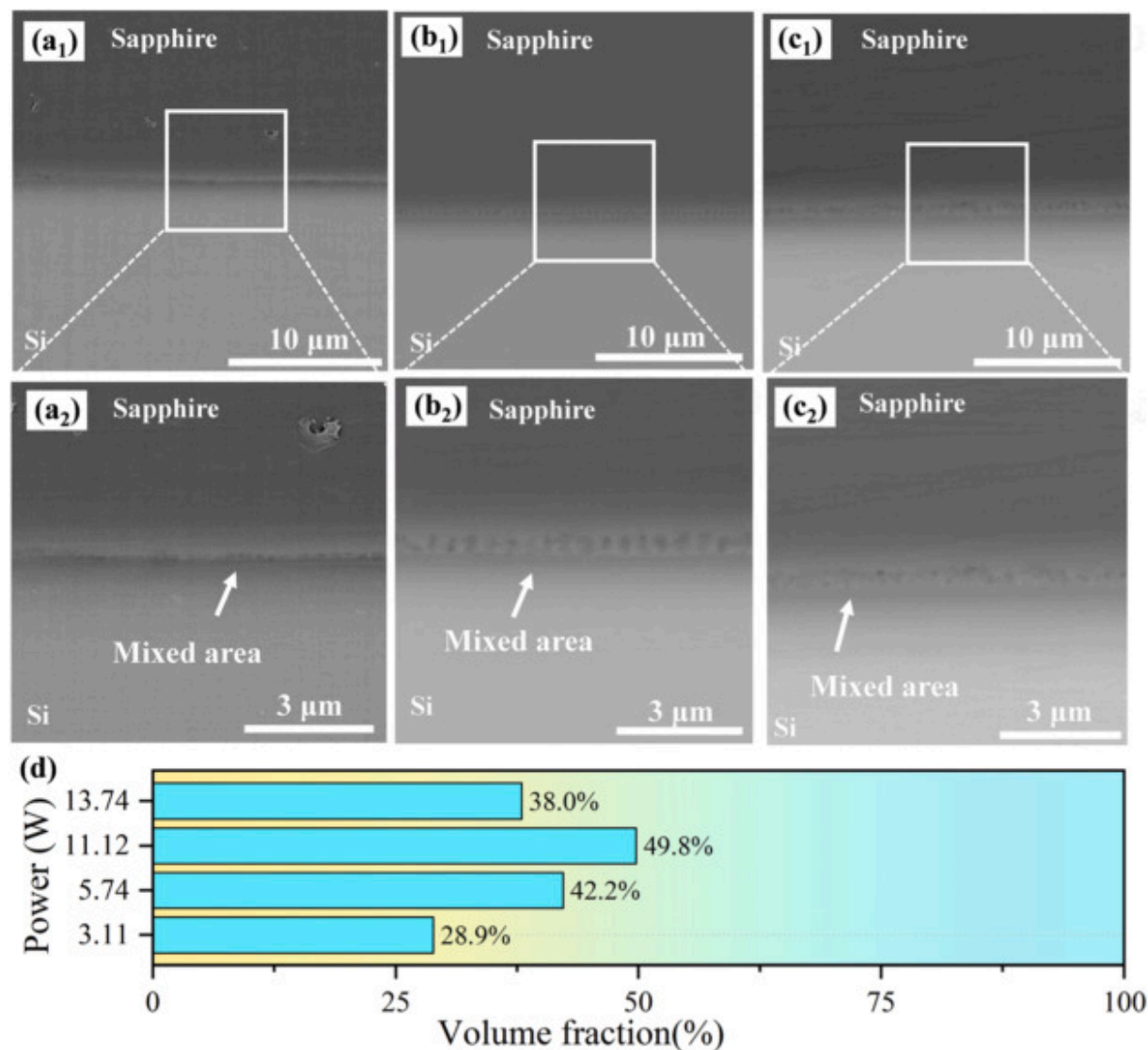
Fig. 7. TEM analysis of sapphire/Si joint:(a) HRTEM image of region II edge; (b) FFT image of region b in Fig. 7(a); (c) SEAD pattern of region c in Fig. 7(a); (d) HRTEM image of region II edge; (e) magnified bright field image of the marked zone e in Fig. 7(d); (f) magnified bright field image of the marked zone f in Fig. 7(d).

In summary, the smooth joint domain comprises three distinct regions, as established by the preceding analyses. Region I corresponds to the mixed area with two phases, including an Al-O-Si amorphous phase and dark Al₂O₃ particles embedded within the Al-O-Si amorphous phase under the impact of the plasma generated by ultrafast laser irradiation at the interface.

Region II exhibits a hybrid structure of locally recrystallized Si and amorphous Si phases. Region III represents the Si substrate, which remains unaffected by the laser irradiation.

3.2. Effect of welding parameters on the microstructure of the sapphire/Si joint

The laser power, scanning speed, and laser pulse width of ultrafast lasers have an important influence on the microstructure of the joint. Fig. 8 illustrates the effect of laser power on the microstructure of the sapphire/Si joint welded by an ultrafast laser with a scanning speed of 50mm/s, a pulse frequency of 200kHz, a pulse width of 800 fs, and a scan spacing of 60μm. As can be seen, at a lower laser power of 3.11 W (peak power density $2.20 \times 10^{14} \text{W/cm}^2$), the interface is well bonded without any microcracks or voids. The thickness of the joint is only around 1μm with negligible elemental interdiffusion due to the limited plasma generation resulting from a small laser energy input. When the laser power increases to 5.74W (peak power density $4.06 \times 10^{14} \text{W/cm}^2$), the microstructure of the joint aligns with the previously described typical structure. The joint comprises Al_2O_3 particles embedded in the Al–O–Si amorphous phase. It is likely attributed to the Si plasma impact generated on the Si substrate during laser irradiation, entraining Al_2O_3 particles into the joint. Concurrently, rapid recrystallization at the joint during cooling embeds these Al_2O_3 particles within the amorphous phases. The joint thickens to ~2.2μm, with no detectable voids, cracks, or bonding defects. When the laser power increases to 11.12W (peak power density $7.87 \times 10^{14} \text{W/cm}^2$), the Al_2O_3 particles grow larger and almost interconnect due to more Si plasma generated from the increased laser energy input. The thickness of the reaction layer in the joint expands to ~2.7μm. When the laser power further increases to 13.74W (peak power density $9.72 \times 10^{14} \text{W/cm}^2$), the amount of Al_2O_3 particles decreases greatly. This is because when the laser power density is increased to a certain level, the thermal deposition at the interface would be sufficient to melt some part of the two base materials at the interface. As a result, fewer Al_2O_3 particles are formed, but more mixing of the two base materials occurs, forming a reaction layer with a thickness of ~3.2μm as the joint domain. Fig. 8(d) shows the volume fraction of $\alpha\text{-Al}_2\text{O}_3$ particles with different laser powers. The volume fraction of $\alpha\text{-Al}_2\text{O}_3$ particles first increases and then decreases with increasing laser power.

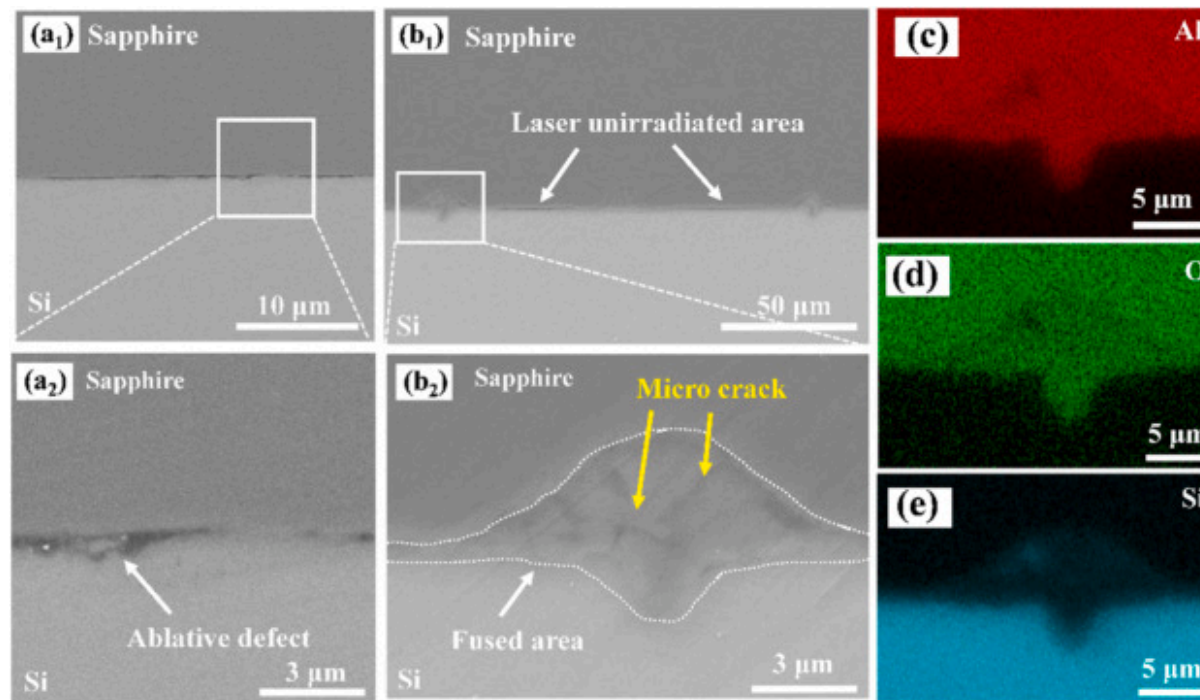


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Fig. 8. Microstructure of sapphire/Si joint at a scanning speed of 50mm/s, pulse frequency of 200kHz, pulse width of 800 fs, and scanning pitch of 60μm:(a₁) - (a₂) 3.11 W; (b₁) - (b₂) 11.12W; (c₁) - (c₂) 13.74W; (d) Volume fraction of α -Al₂O₃ particles with different laser power.

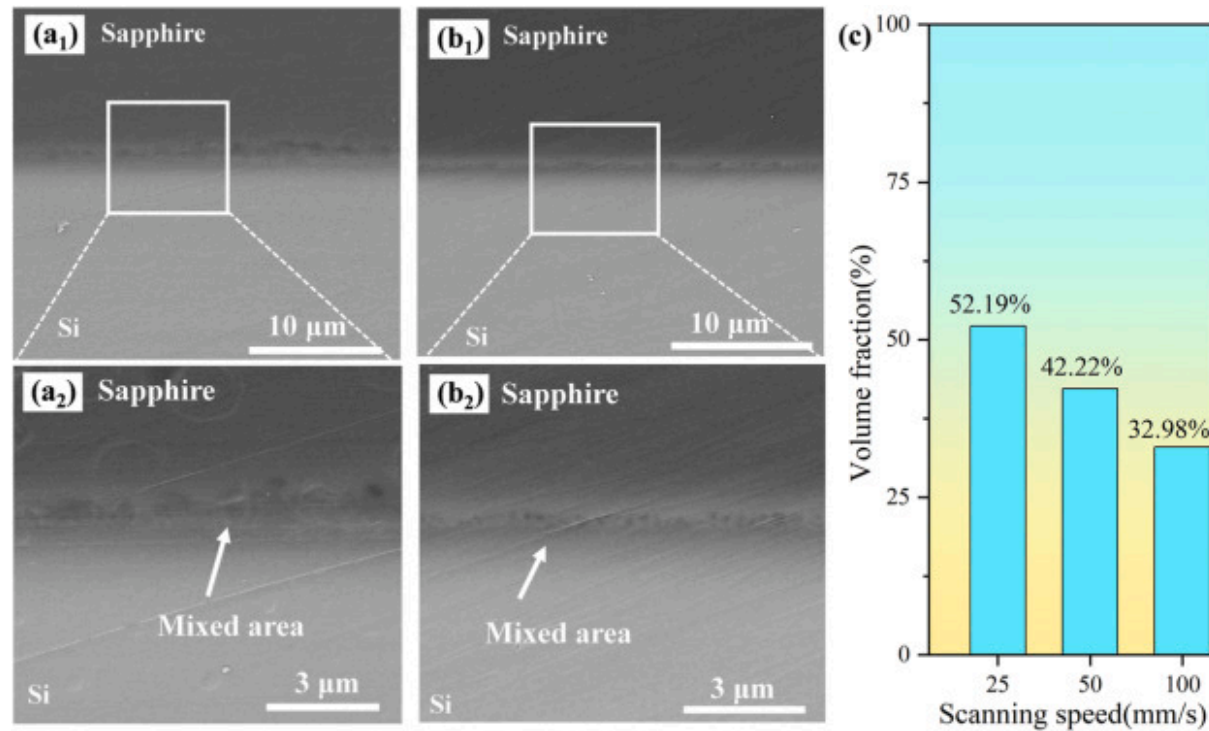
Fig. 9 illustrates the influence of laser pulse width on the microstructure of the sapphire/Si joint. When the laser pulse width is 268 fs, the peak power density is high ($6.69 \times 10^{14} \text{ W/cm}^2$) and the resulting heat effect is relatively low. As a result, the focused ultrafast laser beam causes some ablation defects at the interface and slightly insufficient bonding gaps in the joint due to the insufficient material melting and mixing resulting from the low heat accumulation at the interface, as shown in Fig. 10(a₂). This restricts the thermal diffusion of the sapphire/Si joint and delays the completion of electron-lattice relaxation, thereby hindering thermal equilibrium establishment in the system of the sapphire/Si interface [31].



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Fig. 9. Microstructure of sapphire/Si joint at a laser power of 5.74W, scanning speed of 50mm/s, pulse frequency of 200kHz, and scanning pitch of 60μm: (a₁) -(a₂) 268 fs; (b₁) -(b₂) 1600 fs; (c-f) the EDS elements distribution.



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Fig. 10. Microstructure of sapphire/Si joint at a laser power of 5.74W, pulse frequency of 200kHz, pulse width of 800 fs, and scanning pitch of 60μm: (a₁) -(a₂) 25mm/s; (b₁) -(b₂) 100mm/s; (c) Volume fraction of α-Al₂O₃ particles with different scanning speeds.

When the pulse width is 800 fs, the joint exhibits a smooth characteristic, featuring some Al₂O₃ particles embedded in the joint with no voids or cracks. When the pulse width increases to 1600 fs (peak power density $1.10 \times 10^{14} \text{W/cm}^2$), the joint exhibits a spike-like characteristic. This may be attributed to the more intensive thermal effects, which form the melt pool and facilitate thorough material mixing under the impact of plasma. Due to the increased laser interaction time and the decrease in peak energy density, the lattice could be sufficiently heated. This heating causes the plasma around the interface to expand upward, forming a spike-like joint. As a result, the spike-like joint accumulates more heat than the smooth joint. Compared with the smooth joint with continuous dark matter in Fig. 4, the spike-like joint contains some microcracks in the fused area. EDS

mapping indicates that the enrichment of Si in the spike-like joint at the fused area is significantly higher than in the smooth joint. Due to the insufficient fluidity of the melt pool, unfilled regions are observed in areas unirradiated by the laser. As shown in Fig. 9(c–e), the spike-like joint exhibits a mixed composition of Al, O, and Si, likely resulting in the formation of an amorphous phase. On the one hand, the amorphous structure formed by the intermixing of Al, O, and Si elements was discussed in Section 3.1. On the other hand, the chemical reaction between Si plasma and $\alpha\text{-Al}_2\text{O}_3$ can barely be triggered due to the thermodynamic unfavourability [27].

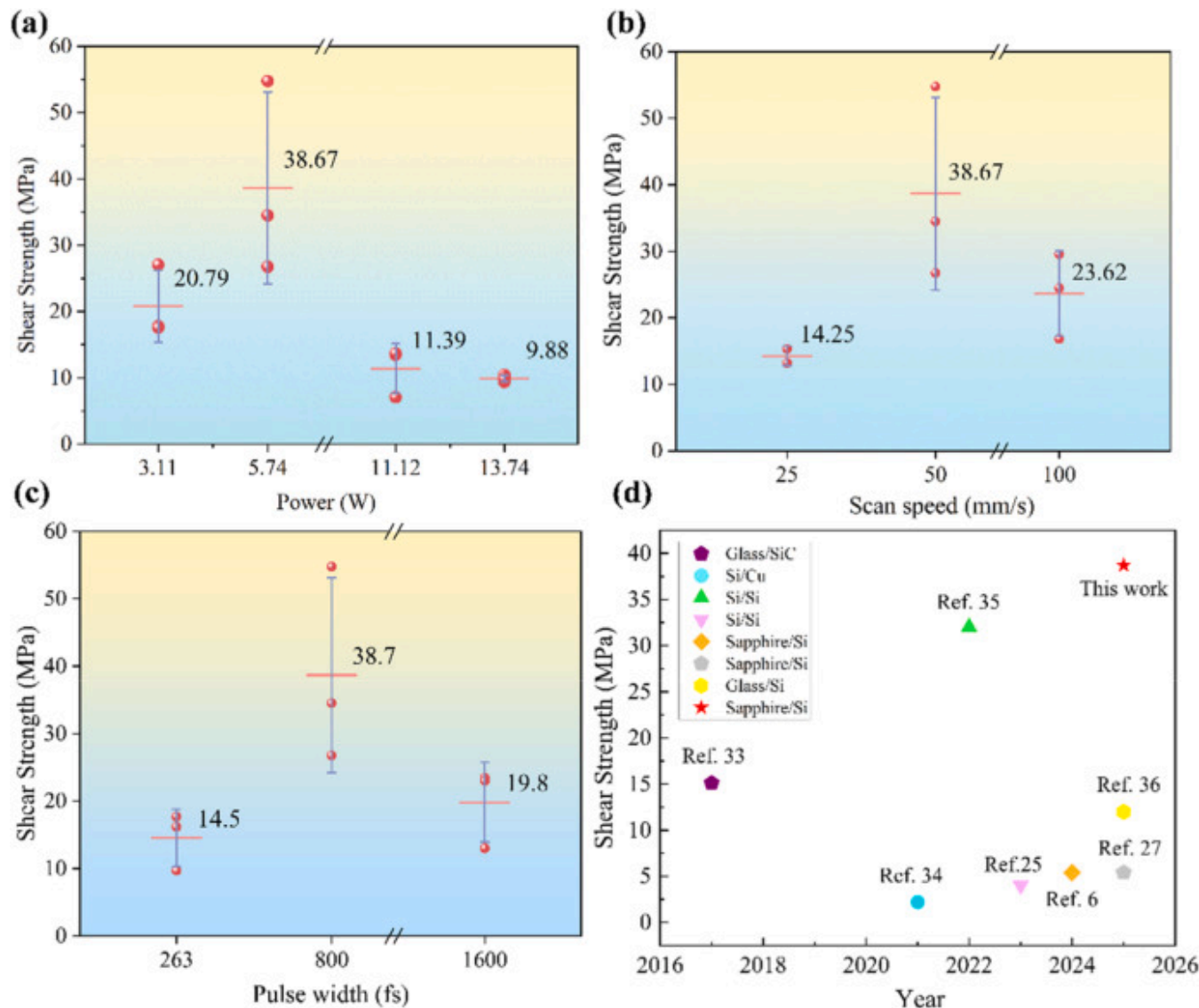
The corresponding microstructure of the sapphire/Si joints with different scanning speeds is shown in Fig. 10. Similar to the influence of laser power variation, scanning speed significantly affects the joint microstructures. When the scanning speed is 25 mm/s (peak power density $4.40 \times 10^{14} \text{ W/cm}^2$), the joint microstructure resembles that observed in Fig. 8(c₁) and (c₂). The reduced scanning speed increases the energy density and thermal deposition, leading to enhanced plasma generation and laser-induced effects. Consequently, Al_2O_3 particles exhibit interconnectivity, and the reaction layer expands to $\sim 3.5 \mu\text{m}$ in the joint. When the scanning speed increases to 50 mm/s (peak power density $2.20 \times 10^{14} \text{ W/cm}^2$), the joint exhibits a smooth characteristic. When the scanning speed is further increased to 100 mm/s (peak power density $1.10 \times 10^{14} \text{ W/cm}^2$), a marked reduction in dark speckle size and reaction layer thickness ($\sim 1.4 \mu\text{m}$) is observed. This phenomenon is attributed to the decreased effective pulse overlap and lower energy input at a higher scanning speed. Induced plasma generation and weaker Si plasma impact result in a limited number of Al_2O_3 particles embedded in the joint. Concurrently, a thinner reaction layer is formed. Fig. 10(c) shows the volume fraction of $\alpha\text{-Al}_2\text{O}_3$ particles with different scanning speeds. The volume fraction of $\alpha\text{-Al}_2\text{O}_3$ particles shows a decreasing trend with increasing scanning speed.

Although different welding processing parameters can yield the same peak power density (at pulse width 1600 fs and scanning speed 100 mm/s), they may produce distinct effects on the microstructure of joints. Fig. 8(c₁) shows the microstructure of the smooth joint, while Fig. 9(b₁) shows the microstructure of the spike-like joint. The joint microstructure exhibits distinct morphologies at the same peak power density ($1.10 \times 10^{14} \text{ W/cm}^2$).

3.3. Effect of ultrafast laser parameters on mechanical properties of the sapphire/Si joint

Shear strength was used to evaluate the mechanical performance of sapphire/Si joints. At least three specimens were tested to determine the average strength value of each experimental condition. Fig. 11(a) shows the shear strength of the sapphire/Si joint with varying laser power. The shear strength first increases and subsequently decreases with rising energy input. When the laser

power is 3.11W, a relatively low shear strength of 20.8MPa is obtained for the joint. It is inferred that the thin reaction and mixed layer in the joint (see Fig. 8(a₂)) results in low interface bonding. When the laser power is 5.74W, the shear strength of the joint peaks at 38.7MPa, which is the highest value among the state-of-the-art Al₂O₃/Si joints (Fig. 11(d), with more detailed information shown in Table 2), to the best knowledge of the authors. It can be seen in Fig. 4 that the joint is composed of continuous Al₂O₃ particles embedded in the amorphous phase, which could prevent the crack propagation within the amorphous region, thus resulting in greater shear capacity. However, large error bars are observed due to stress concentrations caused by irregular dislocations in the transition zone, which generate residual stress at the interface and affect brittle fracture [32]. When the laser power further increases, the decrease in shear strength is attributed to the formation of a thicker reaction layer, which leads to more residual stress.



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Fig. 11. Shear strength of sapphire/Si joint with different welding parameters: (a) Laser power ranging from 3.11 to 13.74W; (b) Laser pulse width ranging from 268 to 1600 fs; (c) Scanning speed ranging from 25 to 100mm/s; (d) The comparison of joint strength between the prepared sapphire/Si joints with several other representative reports.

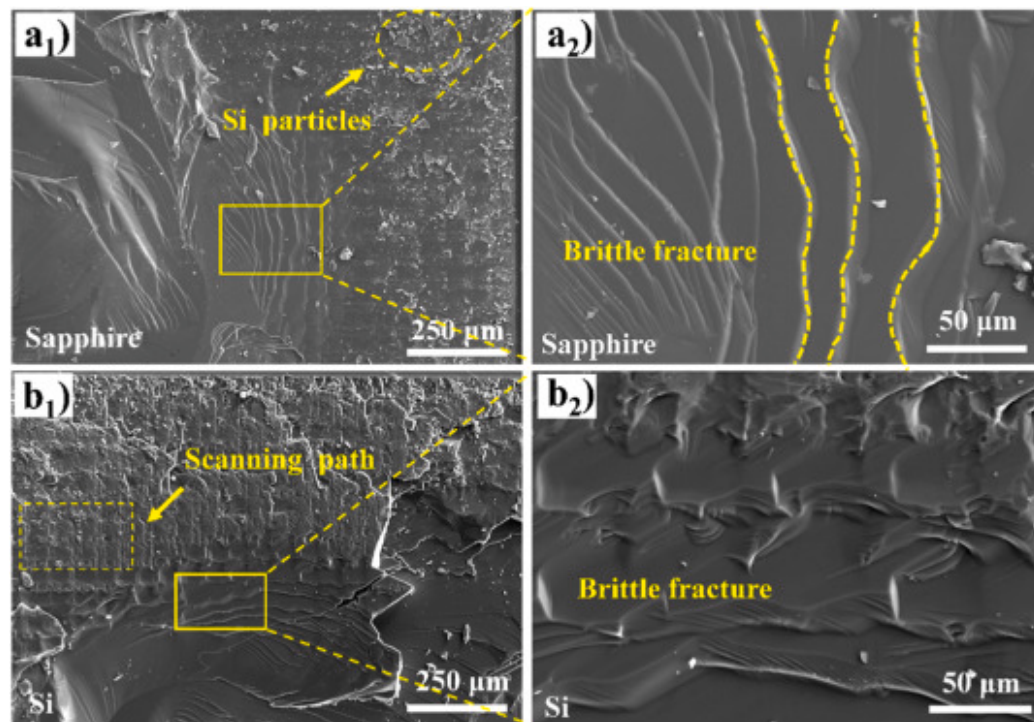
Table 2. The detailed information of several reported semiconductor joints.

Author	Year	Materials	Methods	Strength value	Ref.
G. Zhang et al.	2017	Glass/SiC	Femtosecond laser	15.1 MPa	[33]
M. Chambonneau et al.	2021	Si/Cu	Picosecond Laser	2.2 MPa	[34]
P. Sopeña et al.	2022	Si/Si	Nanosecond laser	32±10 MPa	[35]
M. Chambonneau et al.	2023	Si/Si	Picosecond laser/Gold nanolayers	4 MPa	[25]
Y. Hu et al.	2024	Sapphire/Si	Picosecond laser/Ag NPs	5.4 MPa	[6]
Y. Hu et al.	2025	Sapphire/Si	Femtosecond laser/SiO ₂ layer	5.4 MPa	[27]
X. Zhang et al.	2025	Glass/Si	Picosecond Laser	12 MPa	[36]
R. Pan et al.	2025	Sapphire/Si	Femtosecond laser	38.7 MPa	This work

The effect of laser pulse width on the shear strength of the joint is shown in Fig. 11(b). Similarly, the shear strength first increases and then decreases with the increase of laser pulse width. When the laser pulse width is 268 fs, the ultrashort pulse width leads to ablation and microvoid formation due to insufficient electron-lattice relaxation, resulting in a low shear strength at 15.3 MPa. At 1600 fs, more intensive thermal effects stabilize the melt pool and generate spike-like joints with microcracks, reducing the strength to 22.1 MPa. Meanwhile, the unirradiated regions exhibit weak bonding with limited intermixing, resulting in a relatively low bonding strength of the joint. When the laser pulse width is set at 800 fs in this study, the joint shows good microstructure, and a high shear strength of 38.7 MPa can be obtained.

The effect of laser scanning speed on the shear strength of the joint is shown in Fig. 11(c). As can be seen, consistent with energy density dependencies, a low laser scanning speed of 25 mm/s causes heat accumulation and thermal stress degradation (18.5 MPa). A high laser scanning speed of 100 mm/s reduces the plasma generation and mixing, resulting in a weaker shear strength of 24.7 MPa.

Fig. 12 presents the typical (800 fs, 5.74W, 50mm/s) fracture SEM image of the sapphire/Si smooth joint analyzed in Section 3.1. As shown in Fig. 12(a₁) and (b₁), the fracture propagation path primarily extends along the interface into the Si substrate. In Fig. 12(a₁), residual Si particles can be observed on the Al₂O₃ fracture surface. Fig. 12(b₁) shows the scanning path of cross lines on the Si fracture surface, demonstrating that some fractures occur at the interface of the joint. As shown in Fig. 12(a₂) and (b₂), the fracture surface exhibits clear characteristics of brittle fracture, and also occurs near the Si substrate of the interface. According to the TEM analysis in Section 3.1, the brittle fractures may be attributed to stress concentrations caused by irregular dislocations in the transition region between the joint and the Si substrate.



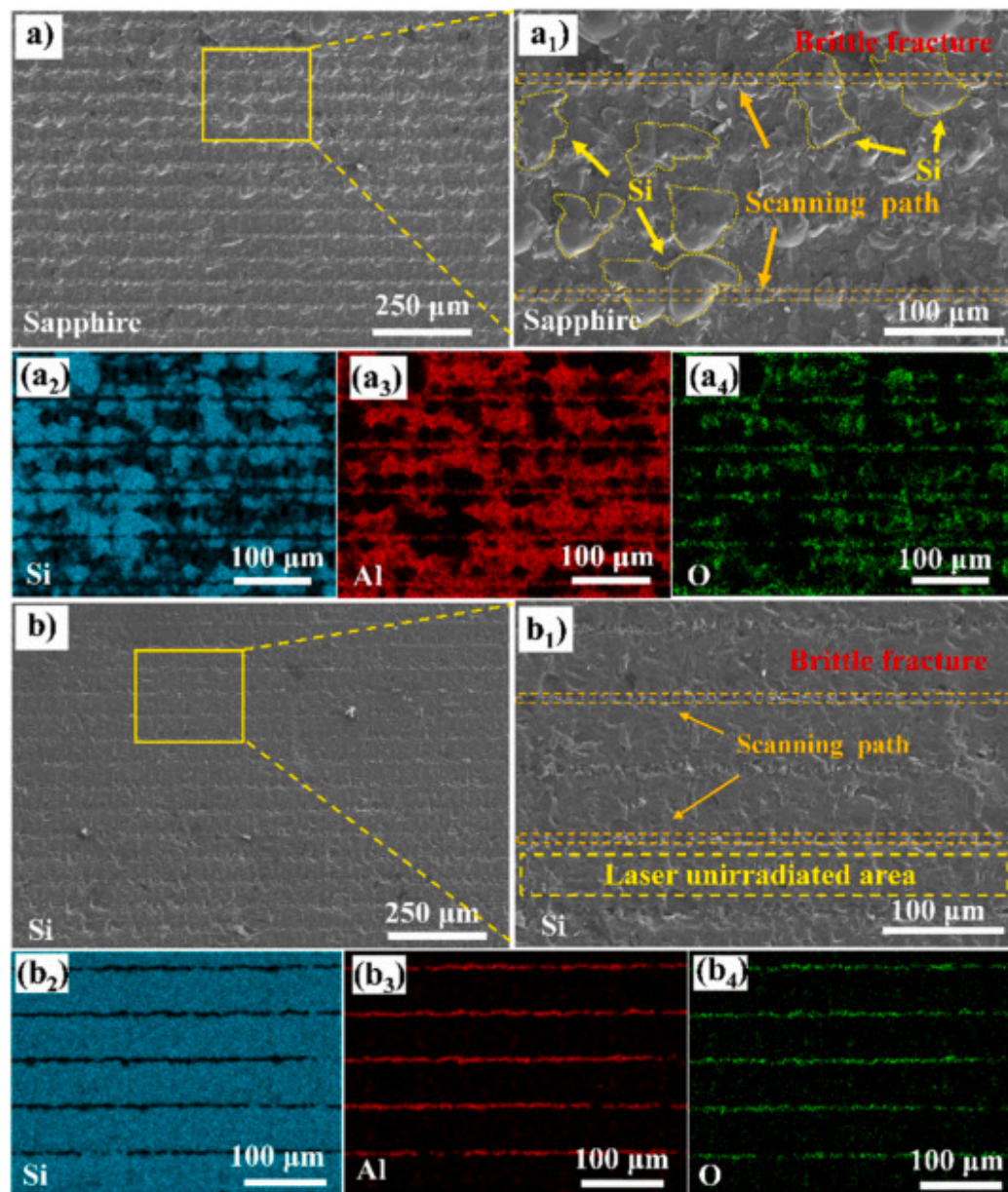
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Fig. 12. SEM image of the fracture surface of the sapphire/Si smooth joint:(a₁) - (a₂) at the sapphire side (b₁) - (b₂) at the Si side.

Fig. 13 shows the brittle fracture surface and corresponding EDS mapping of the spike-like sapphire/Si joint. As shown in Fig. 13(a₁) and (a₂), a large amount of Si can be observed adhering to the sapphire side. Fig. 13(b₁) shows that fracture propagation

follows the laser scanning path, while the laser-unirradiated areas remain smooth. The EDS mapping in Fig. 13(b₂-b₄) indicates that the scanning path consists of Al and O elements, contrasting with the elemental Si dominated in the unirradiated areas. This fracture morphology confirms that the fracture originated from the joint of the sapphire/Si interface.

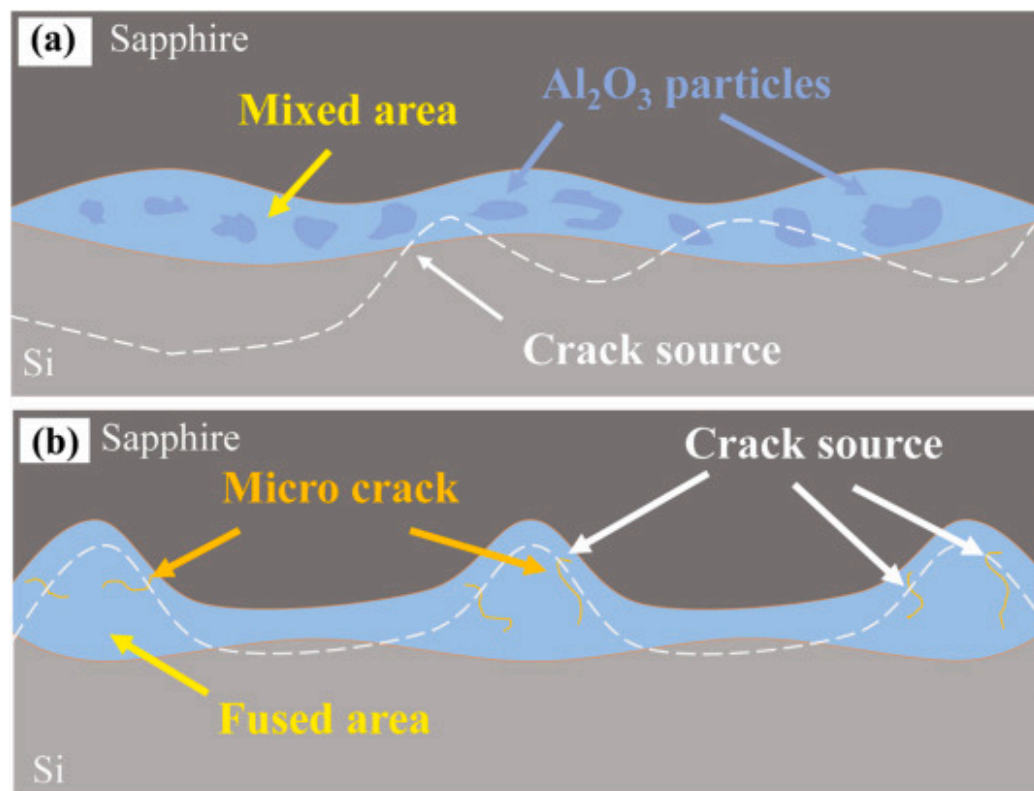


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Fig. 13. SEM image of the fracture surface of the sapphire/Si spike-like joint:(a) - (a₄) at the sapphire side and correspondingly Al, O, and Si elemental mapping (b₁) - (b₂) at the Si side and correspondingly Al, O, and Si elemental mapping.

According to the above fracture surface analysis, the fracture profiles of the joint can typically be divided into two categories. Fig. 14(a) shows the Type I fracture mechanism in smooth joints. These nanoscale Al₂O₃ particles can impede crack propagation within the amorphous region during the fracture process, thereby enhancing the toughness of the joint [37,38]. As a result, the resulting shear strength is higher than that reported in existing studies. When the energy input increases further, the Al₂O₃ particles become larger and the reaction layer thickens, reducing the crack-barrier effect of the Al₂O₃ particles and consequently decreasing the toughness of the joint. Meanwhile, stress concentrations generated from irregular dislocations in the transition region also possibly lead to fracture, with crack propagation primarily extending along the interface into the Si substrate. Fig. 14(b) indicates the Type II fracture mechanism observed in spike-like joints. Compared to the smooth joint, the spike-like joint primarily bears the shear load. However, the unirradiated regions exhibit lower resistance, which leads to a reduction in the overall shear strength of the joint. This fracture mode occurs primarily within the joint and partly extends into the adjacent Si substrate. Based on the characteristic morphology of the spike-like sapphire/Si joint, it is inferred that the presence of microcracks within the joint serves as a crack source. The residual silicon adhering to the sapphire side may be due to the larger laser pulse width, which produces heat accumulation and thermal stress.



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Fig. 14. Schematic of different fracture modes of sapphire/Si joint: (a) Fracture mode I of smooth joint; (b) Fracture mode II of spike-like joint.

4. Conclusion

In summary, this study successfully achieved direct joining of sapphire/Si joints using ultrafast laser micro-welding. The joint microstructure was systematically characterized, and the influences of process parameters on joint formation were thoroughly investigated. The conclusions are as follows:

1. The interfaces of the sapphire/Si joint are well bonded without voids or microcracks, verifying the feasibility of ultrafast laser welding of sapphire and Si. The smooth sapphire/Si joint is mainly composed of Al-O-Si amorphous phases, and Al₂O₃ particles embedded in the amorphous phase can be observed.
2. Laser parameters critically affect joint microstructure and mechanical performance. At a pulse width of 800 fs, smooth joints can be obtained by appropriately balancing laser power and scanning speed. Low laser power and high scanning speed lead to insufficient Si plasma and poor interfacial mixing features. High laser power and low scanning speed could cause thermal accumulation, excessive stress, and reduced shear strength. A short pulse width (268 fs) leads to interfacial ablation and micro-void formation due to insufficient electron-lattice relaxation. A long pulse width (1600 fs) enhances severe thermal effects and forms spike-like joints containing microcracks.
3. By optimizing the laser welding parameters, defect-free sapphire/Si joints with a maximum shear strength of 38.7MPa can be achieved using 5.74W laser power, 100mm/s scanning speed, and a pulse width of 800 fs. Brittle fracture is observed in all joints, and two corresponding fracture models are established. In smooth joints, brittle failure occurs within the region consisting of recrystallized Si and amorphous Si phases, whereas in spike-like joints, brittle failure originates from the microcracks formed within the joint region.

CRediT authorship contribution statement

Rui Pan: Writing – original draft, Conceptualization. **Bingxu Guo:** Writing – original draft, Investigation, Data curation. **Jiayi Xu:** Validation, Methodology, Investigation. **Fuqiang Ye:** Validation, Investigation. **Yao Du:** Resources, Investigation. **Yinghao Feng:** Visualization, Resources. **Pei Chen:** Supervision, Project administration. **Shujun Chen:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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[Recommended articles](#)

References

- [1] N.J.R. Muniraj
MEMS based humidity sensor using Si cantilever beam for harsh environmental conditions
Microsyst. Technol., 17 (2011), pp. 27-29
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [2] D. Yang, J. Chen, X. Ma, D. Que
Impurity engineering of Czochralski silicon used for ultra large-scaled-integrated circuits
J. Cryst. Growth, 311 (2009), pp. 837-841
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [3] R.R.A. Syms, D.F. Moore
Optical MEMS for telecoms
Mater. Today, 5 (2002), pp. 26-35
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [4] D.C. Harris
Durable 3–5 μm transmitting infrared window materials
Infrared Phys. Technol., 39 (1998), pp. 185-201
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [5] A. de Pablos-Martín, S. Tismer, T. Höche
Structural characterization of laser bonded sapphire wafers using a titanium absorber thin film

J. Mater. Sci. Technol., 31 (2015), pp. 484-488



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [6] Y. Hu, L. Lin, J. Xu, S. Hu, J. Ji, W. Wu, X. Zuo, Q. Zhang, Z. Li
Metal nanoparticles assisted ultrafast laser plasmonic microwelding of oxide–semiconductor interconnects
Small Methods, 8 (2024), Article 2301232
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [7] R. Pan, Q. Wang, D. Sun, P. He
Effects of electric field on interfacial microstructure and shear strength of diffusion bonded α -Al₂O₃/Ti joints
J. Eur. Ceram. Soc., 35 (2015), pp. 219-226
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [8] T. Matsumae, Y. Kurashima, H. Umezawa, Y. Mokuno, H. Takagi
Room-temperature bonding of single-crystal diamond and Si using Au/Au atomic diffusion bonding in atmospheric air
Microelectron. Eng., 195 (2018), pp. 68-73
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [9] W. Guo, L. Fu, T. Lin, P. He, C. Wang, T. Wang, H. Liu
New design of sapphire joints brazed with bismuth–borate glass
Ceram. Int., 45 (2019), pp. 5036-5049
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [10] S. Zhang, Y. Zhao, R. Xu, W. Wang, X. Li, X. Zhang, B. Guo, P. Lin, T. Lin, S. Chen, R. Pan
Enhancing sapphire/Kovar alloy brazed joint through combined effect of TiB whisker reinforcement and surface micro-nano structures
Mater. Char., 217 (2024), Article 114439
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [11] P. Montméat, T. Enot, M. De Marco Dutra, M. Pellat, F. Fournel

Study of a silicon/glass bonded structure with a UV-curable adhesive for temporary bonding applications

Microelectron. Eng., 173 (2017), pp. 13-21



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[12]

I.E. Tyschenko, E.D. Zhanaev, V.P. Popov

Bonding energy of silicon and sapphire wafers at elevated temperatures of joining

Semiconductors, 53 (2019), pp. 60-64

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[13]

S. Murakami, R. Takigawa

Investigation of Si/Si bond interface fabricated using room temperature direct bonding

Surf. Interfaces, 66 (2025), Article 106124



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[14]

F. Niu, S. Yang, X. Yuan, Q. Kang, L. Liu, C. Wang

Plasma activated titanium-based bonding for robust and reliable Si-Si and Si-glass integration

Ceram. Int., 50 (2024), pp. 25182-25191



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[15]

R.R. Gattass, E. Mazur

Femtosecond laser micromachining in transparent materials

Nat. Photonics, 2 (2008), pp. 219-225

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[16]

J. Xianshi, L. Jinlin, L. Kai, W. Cong, L. Zhou, W. Mengmeng, J. Zhengyi, P.V. Vadim, D. Ji'an

Ultrafast laser welding of transparent materials: from principles to applications

Int. J. Extrem. Manuf., 7 (2025), Article 032001

[Google Scholar ↗](#)

[17]

T. Tamaki, W. Watanabe, K. Itoh

Laser micro-welding of transparent materials by a localized heat accumulation effect using a femtosecond fiber laser at 1558 nm

Opt. Express, 14 (2006), pp. 10460-10468

[View in Scopus ↗](#) [Google Scholar ↗](#)

[18] J. Huo, Z. Zeng, J. Yuan, M. Luo, A. Luo, J. Li, H. Yang, N. Zhao, Q. Zhang

Welding between rough copper foil and silica glass using green femtosecond laser

Opt Laser. Technol., 181 (2025), Article 111804

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[19] C. Tian, H. Ren, H. Shen

The connection of glass and metal with a large gap by combining laser soldering and ultrafast laser welding

J. Manuf. Process., 102 (2023), pp. 528-534

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[20] R. Pan, D. Yang, T. Zhou, Y. Feng, Z. Dong, Z. Yan, P. Li, J. Yang, S. Chen

Micro-welding of sapphire and metal by femtosecond laser

Ceram. Int., 49 (2023), pp. 21384-21392

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[21] X. Zuo, L. Lin, Y. Hu, J. Ji, W. Wu, Q. Zhang, Z. Li

High-strength and impermeable sapphire/aluminum joints fabricated by ultrafast laser microwelding: microstructures and joining mechanism

Opt Laser. Technol., 180 (2025), Article 111455

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[22] H. Yuan, C. Li, X. Si, B. Yang, M. Li, W. Zhao, D. Gao, J. Cao

Ultrashort pulse laser welding of alumina ceramic and titanium

J. Eur. Ceram. Soc., 44 (2024), pp. 3393-3399

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [23] E.H. Penilla, L.F. Devia-Cruz, A.T. Wieg, P. Martinez-Torres, N. Cuando-Espitia, P. Sellappan, Y. Kodera, G. Aguilar, J.E. Garay
Ultrafast laser welding of ceramics
Science, 365 (2019), pp. 803-808
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [24] S. Zhang, X. Li, F. Lu, J. Xu, R. Xu, X. Zhang, Y. Zhao, X. Sun, R. Pan, P. Lin, T. Lin, S. Chen, P. He
Femtosecond laser-induced honeycomb structure on the interface for the micro-welding of YSZ/Sapphire
J. Adv. Ceram. 14 (2025), p. 9221052
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [25] M. Chambonneau, Q. Li, M. Blothe, S.V. Arumugam, S. Nolte
Ultrafast laser welding of silicon
Advanced Photonics Research, 4 (2023), Article 2200300
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [26] I. Miyamoto, Y. Okamoto, A. Hansen, J. Vihinen, T. Amberla, J. Kangastupa
High speed, high strength microwelding of Si/glass using ps-laser pulses
Opt. Express, 23 (2015), pp. 3427-3439
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [27] Y. Hu, L. Lin, J. Ji, W. Wu, X. Zuo, Z. Cai, H. Li, H. Shang, Z. Li
Chemical-resistant, highly-impermeable integration of large differential semiconductor and oxide by spatial-confined plasma assisted ultrafast laser microwelding for optofluidic microsystem
Small (2025), Article 2500881
n/a
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [28] H. Chen, Y. Rong, Y. Huang, C. Wu
Crack suppression of glass welding by ultrafast laser without optical contact based on light modulation
Opt Laser. Technol., 164 (2023), Article 109466
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [29] S. Richter, F. Zimmermann, R. Eberhardt, A. Tünnermann, S. Nolte
Toward laser welding of glasses without optical contacting
Appl. Phys. A, 121 (2015), pp. 1-9
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [30] Y. Feng, R. Pan, T. Zhou, Z. Dong, Z. Yan, Y. Wang, P. Chen, S. Chen
Direct joining of quartz glass and copper by nanosecond laser
Ceram. Int., 49 (2023), pp. 36056-36070
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [31] B.N. Chichkov, C. Momma, S. Nolte, F. von Alvensleben, A. Tünnermann
Femtosecond, picosecond and nanosecond laser ablation of solids
Appl. Phys. A, 63 (1996), pp. 109-115
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [32] H. Moshayedi, I. Sattari-Far
The effect of welding residual stresses on brittle fracture in an internal surface cracked pipe
Int. J. Pres. Ves. Pip., 126–127 (2015), pp. 29-36
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [33] G. Zhang, J. Bai, W. Zhao, K. Zhou, G. Cheng
Interface modification based ultrashort laser microwelding between SiC and fused silica
Opt. Express, 25 (2017), pp. 1702-1709
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [34] M. Chambonneau, Q. Li, V.Y. Fedorov, M. Blothe, K. Schaarschmidt, M. Lorenz, S. Tzortzakis, S. Nolte
Taming ultrafast laser filaments for optimized semiconductor–metal welding
Laser Photon. Rev., 15 (2021), Article 2000433
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [35] P. Sopeña, A. Wang, A. Mouskeftaras, D. Grojo

Transmission laser welding of similar and dissimilar semiconductor materials

Laser Photon. Rev., 16 (2022)

[Google Scholar ↗](#)

- [36] X. Zhang, Y. Zhao, M. Wu, M. Yang, T. Zhang, L. Zhang, J. Yang, C. Tan, X. Song
Ultrafast laser packaging of glass and silicon: microstructure and mechanical property
Opt Laser. Technol., 184 (2025), Article 112538



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [37] W.W. Gerberich, J. Michler, W.M. Mook, R. Ghisleni, F. Östlund, D.D. Stauffer, R. Ballarini
Scale effects for strength, ductility, and toughness in “brittle” materials
J. Mater. Res., 24 (2009), pp. 898-906

[Crossref ↗](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [38] P. Garg, T.J. Rupert
Grain incompatibility determines the local structure of amorphous grain boundary complexes
Acta Mater., 244 (2023), Article 118599



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

Cited by (0)

[View Abstract](#)

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